

Protobuf / Connect Service Spec

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Master technical specification — Volume 3 (Control Plane, Go), nano-detail document. Scope: the **complete Coordinator service contract** — the single `.proto` that every HelixVPN agent (Access client, Connector, edge) speaks, served by the Go control plane over **Connect** (gRPC / gRPC-Web / Connect protocol on one listener). This document deepens the "Agent contract — protobuf" part of [02-control-plane.md §4] into an implementation-ready, field-by-field, wire-by-wire specification. It is a SPEC: it describes the contract precisely enough to generate code and tests; it does not build the product.

Source evidence cited inline by id: [04_P1] = 04_VPN_CLD/HelixVPN-Phase1-MVP.md, [04_ARCH §N] = 04_VPN_CLD/HelixVPN-Architecture-Refined.md, [research-go_cp] = final/02-control-plane.md, [SYNTHESIS §N] = the cross-document synthesis. Unproven / not-yet-decided facts are marked **UNVERIFIED** per constitution §11.4.6 — never fabricated.

0. What this document owns, and the one reconciliation it makes

0.1 Ownership boundary

This document owns the **canonical .proto** for the agent $\rightleftarrows$ control-plane contract and its **server-side semantics**: every service method, every message, every field number, every field's meaning, the snapshot/delta/keepalive state machine, the `known_version` cheap-reconnect protocol, the policy-prefiltering guarantee, the error taxonomy, the authz rules at the RPC boundary, the convergence SLO the contract must meet, and the buf codegen pipeline that emits Go/Dart/Rust stubs.

It does **not** own: the byte-level evolution/back-compat rules of `WatchNetworkMap` across versions (that is the dedicated `WatchNetworkMap` contract document in this same Volume 3); the Go module wiring, DDL, RLS, IPAM, policy compiler internals, or Redis Streams envelope (those are [research-go_cp §1–§7]); the Rust data plane (Volume 1); the client UI (Volume 5). Where this document references those, it cites them and does not redefine them.

0.2 Package-name reconciliation (the one correction this document makes)

[research-go_cp §4] declares the proto with a **package/path mismatch**: the package line reads `package helix.coordinator.v1`; while the file is placed at `proto/helix/coordinator/v1/coordinator.proto` with option `go_package = ".../gen/helix/coordinator/v1;coordinatorv1"`. Per the Volume-3 authoring directive, the **unified package across the whole HelixVPN set is `helix.coordinator.v1`**. This document therefore makes the file path, the `go_package`, and the generated-stub identifiers agree with that package, so the four surfaces (package, file path, Go import path, Dart/Rust package) cannot drift:

Surface	Canonical value (this spec)	Was in [research-go_cp §4]
proto package	<code>helix.coordinator.v1</code>	<code>helix.coordinator.v1</code> ✓
file path	<code>proto/helix/coordinator/v1/coordinator.proto</code>	<code>proto/helix/coordinator/v1/coordinator.proto</code> ✕
option <code>go_package</code>	<code>github.com/vasic-digital/helix-go/gen/helix/coordinator/v1;coordinatorv1</code>	<code>.../gen/helix/coordinator/v1;coordinatorv1</code> ✕
Go stub alias (this spec)	<code>coordinatorv1</code>	<code>coordinatorv1</code>

This is a visible reconciliation per constitution §11.4.35 (no silent rename): the **wire identity** (`helix.coordinator.v1.Coordinator`) was already correct in the source; only the file layout and Go identifiers are brought into line. All Go sketches in [research-go_cp] that import `coordinatorv1` are understood to mean `coordinatorv1` after this reconciliation; their **behaviour is unchanged** because protobuf wire compatibility keys off the package + message name, not the file path or the Go alias.

1. The complete .proto (canonical, normative)

This is the **single source of truth** for the agent contract. Every field number is fixed for the life of v1 and MUST NOT be reused or renumbered (proto3 wire rule; enforced by `buf breaking`, §6.3). Comments are normative where they state a guarantee (C-rules from [research-go_cp §0.1]).

```
// proto/helix/coordinator/v1/coordinator.proto
//
// HelixVPN agent-control-plane contract. Served over Connect
//   (gRPC / gRPC-Web / Connect).
// Canonical package across the whole HelixVPN set:
//   helix.coordinator.v1 (§0.2).
syntax = "proto3";

package helix.coordinator.v1;
```

```

option go_package = "github.com/vasic-digital/helix-go/gen/helix/
    coordinator/v1;coordinatorv1";

// -----
// Service
// -----
service Coordinator {
    // One-time device enrollment: exchange a single-use enroll
    // token for a durable
    // identity (device_id + overlay_ip) and a short-lived control-
    // channel mTLS cert.
    // The ONLY unauthenticated RPC (it authenticates by the hashed
    // enroll token, §4.1).
    rpc Enroll(EnrollRequest) returns (EnrollResponse);

    // The spine of the system. Open once over the device mTLS
    // channel; receive a
    // snapshot (or resume-deltas, §3.3), then an open-ended delta
    // stream until the
    // agent disconnects. Push, don't poll (C5). Convergence event-
    // >delta p99 < 1s (§5).
    rpc WatchNetworkMap(WatchRequest) returns (stream MapUpdate);

    // Connectors push the CIDRs they serve. Idempotent: re-sending
    // the same set is a
    // no-op. Also settable out-of-band via Console/REST (POST /
    // v1/.../prefixes).
    rpc AdvertisePrefixes(AdvertiseRequest) returns
        (AdvertiseResponse);

    // Lightweight presence/health heartbeat. Carries ONLY transport
    // + rtt. Carries NO
    // bytes / flows / destinations / packet counts – no-logging by
    // construction (C3).
    rpc ReportStatus(StatusReport) returns (StatusAck);
}

// -----
// Enrollment
// -----
message EnrollRequest {
    string enroll_token =
        1; // single-use, short-lived; server stores only its
        hash (§4.1)
    bytes wg_pubkey = 2; // 32-byte Curve25519 PUBLIC key;
        private key never leaves device (C6)
    string os = 3; //
        "ios"|"android"|"linux"|"windows"|"macos"|"harmonyos"|"aurora"
    string name = 4; // human label, e.g. "alice-
        pixel-8"
    DeviceKind kind = 5; // CLIENT or CONNECTOR
}

enum DeviceKind {

```

```

    DEVICE_KIND_UNSPECIFIED = 0; // proto3 zero value; an
        EnrollRequest with this kind is rejected (§4.6)
    CLIENT = 1;
    CONNECTOR = 2;
}

message EnrollResponse {
    string device_id = 1; // server-assigned UUID
        (stringified)
    string overlay_ip = 2; // IPAM-allocated, e.g.
        "fd7a:helix:1::2" – tenant ULA /48 (D4)
    bytes device_cert = 3; // DER mTLS client cert for the
        control channel (~24h TTL)
    GatewayInfo gateway =
        4; // where to open WatchNetworkMap + data-plane endpoint
}

// -----
// WatchNetworkMap
// -----

message WatchRequest {
    string device_id =
        1; // MUST equal the device bound to the mTLS cert
        (§4.4); else PermissionDenied
    uint64 known_version =
        2; // 0 => full snapshot; else => resume with deltas
        since this version (§3.3)
}

// Every stream frame. Exactly one body variant is set per frame.
message MapUpdate {
    uint64 version = 1; // monotonic per tenant; the version
        this frame brings the agent TO
    oneof body {
        NetworkMap snapshot = 2; // full desired state (sent first,
            or on forced resync)
        MapDelta delta = 3; // incremental change since the
            agent's prior version
        KeepAlive keepalive = 4; // liveness only; carries no state
            (version unchanged)
    }
}

message NetworkMap {
    Node self = 1; // this agent's own identity
        within the overlay
    GatewayInfo gateway = 2; // current gateway/relay
        endpoint + key
    repeated Peer peers = 3; // peers this node MAY reach –
        ALREADY policy-filtered (C4)
    DnsConfig dns =
        4; // overlay DNS the client should use
    TransportPolicy transport =
        5; // obfuscation escalation ladder (doc-01 walks it)
}

```

```

message Node {
    string device_id = 1;
    string overlay_ip = 2; // this node's own ULA address
}

message GatewayInfo {
    string endpoint =
        1; // "gw.example:443" - host:port the data plane dials
    bytes wg_pubkey = 2; // gateway WG public key (32-byte
        Curve25519)
    string masque_sni =
        3; // SNI/host to present for MASQUE/HTTP-3 masquerade
        (Volume 1, D1)
}

message Peer {
    string device_id = 1;
    bytes wg_pubkey = 2; // 32-byte Curve25519
        public key
    repeated string allowed_ips =
        3; // compiled WG AllowedIPs - ONLY what this node may
        reach (C4)
    string endpoint = 4; // MVP: gateway relay
        endpoint; P2 adds direct candidates
    bool is_connector = 5; // true => this peer
        advertises LAN prefixes
    repeated Via6Route via6 = 6; // 4via6 mappings for
        colliding IPv4 LANs (D4)
}

// 4via6: a deterministic IPv6 encoding of a colliding IPv4 CIDR
// so the client can
// always disambiguate which "192.168.1.0/24" it means (D4,
// [04_ARCH §3.4]).
message Via6Route {
    string ipv4_cidr = 1; // e.g. "192.168.1.0/24"
    string via6_prefix = 2; // e.g. "fd7a:helix:1:5::/96" - site-
        scoped IPv6 mapping
}

message DnsConfig {
    repeated string resolvers = 1; // overlay DNS resolver
        addresses
    repeated string search = 2; // DNS search domains
}

message TransportPolicy {
    repeated string order = 1; // ordered ladder,
        e.g. ["plain-udp", "lwo", "masque-h3"]
    bool allow_user_override = 2; // may the end user
        pin a transport in the UI?
}

message MapDelta {

```

```

    repeated Peer    upsert_peers    = 1; // add-or-replace these
                                   peers (match by device_id)
    repeated string  remove_peer_ids = 2; // device_ids the agent
                                   MUST drop (revoke / lost visibility)
    TransportPolicy transport        =
        3; // PRESENT ONLY IF CHANGED (else unset => unchanged)
    DnsConfig        dns             =
        4; // PRESENT ONLY IF CHANGED (else unset => unchanged)
}

// -----
// AdvertisePrefixes (connectors only)
// -----
message AdvertiseRequest {
    string    device_id = 1; // MUST be a CONNECTOR bound to
                           the mTLS cert (§4.4)
    repeated string cidrs    = 2; // the COMPLETE set the
                           connector serves (declarative, not a diff)
}

message AdvertiseResponse {
    bool    accepted = 1; // true => stored; false => see
                       conflicts
    repeated string conflicts = 2; // human-readable overlap/
                       validation messages (§4.5)
}

// -----
// ReportStatus (presence + health ONLY – C3)
// -----
message StatusReport {
    string device_id = 1;
    string transport =
        2; // currently-active transport label, e.g. "masque-h3"
        (health, not traffic)
    uint32 rtt_ms    = 3; // round-trip estimate in milliseconds
        (health, not traffic)
    // INTENTIONALLY ABSENT: bytes, flows, destinations, packet
        counts, SNI, timestamps-per-packet.
}

message StatusAck {} // empty: acknowledgement only
message KeepAlive {} // empty: liveness frame body

```

1.1 Field-number registry (frozen for v1)

Field numbers are the wire identity; they are listed here so a reviewer can confirm none is reused. Reserving a removed number (proto3 reserved) is mandatory if any field is ever deleted; no v1 field is deleted in this revision, so there are no reserved ranges yet.

Message	#	Field	Type	Notes
EnrollRequest	1	enroll_token	string	

Message	#	Field	Type	Notes
				hashed server-side
	2	wg_pubkey	bytes(32)	public only (C6)
	3	os	string	enum-shaped string (see §4.6)
	4	name	string	
	5	kind	DeviceKind	UNSPECIFIED rejected
EnrollResponse	1	device_id	string	
	2	overlay_ip	string	ULA /48
	3	device_cert	bytes	DER
	4	gateway	GatewayInfo	
WatchRequest	1	device_id	string	must match cert
	2	known_version	uint64	0 = snapshot
MapUpdate	1	version	uint64	monotonic/tenant
NetworkMap	2/3/4	snapshot/delta/keepalive	oneof body	exactly one set
	1	self	Node	
	2	gateway	GatewayInfo	
	3	peers	repeated Peer	policy-filtered (C4)
	4	dns	DnsConfig	
Peer	1..6	transport	TransportPolicy	
		device_id, wg_pubkey, allowed_ips, endpoint, is_connector, via6	mixed	
MapDelta	1..4	upsert_peers, remove_peer_ids, transport, dns	mixed	3/4 optional-if-changed
AdvertiseRequest	1/2	device_id, cidrs	string / repeated string	declarative set
AdvertiseResponse	1/2	accepted, conflicts	bool / repeated string	
StatusReport	1/2/3	device_id, transport, rtt_ms	string/string/uint32	NO traffic fields (C3)

2. Connect transport binding

The single Coordinator service is served over **Connect** (the `connectrpc.com/connect` runtime) on the same HTTP listener as the Gin REST API [research-go_cp §8]. Connect makes one service simultaneously addressable three ways, so native agents and a future WASM Console use the **same** generated stubs [04_P1 §3]:

Caller	Wire protocol	HTTP version	Used by
Native agent (Rust/Go/Dart FFI)	gRPC	HTTP/2	Access client, Connector, edge
Browser	gRPC-Web	HTTP/1.1 or /2	WASM Console (Phase 2/3)
Generic HTTP client	Connect protocol	HTTP/1.1	curl-style ops, debugging

flowchart LR

subgraph Agents

A[Access client] -->|gRPC/HTTP2 + device mTLS| L

C[Connector] -->|gRPC/HTTP2 + device mTLS| L

E[Edge] -->|gRPC/HTTP2 + device mTLS| L

end

B[WASM Console] -->|gRPC-Web/HTTP1.1| L

L[(Single HTTPS listener :8443)] --> C0[Connect Coordinator handler]

L --> GIN[Gin REST/WS handler]

C0 --> COORD[internal/coordinator]

C0 --> ID[internal/identity]

C0 --> REG[internal/registry]

Binding rules.

- WatchNetworkMap is a **server-streaming** RPC → Connect ServerStream[MapUpdate]. It **MUST** run over HTTP/2 for native agents (HTTP/1.1 server-streaming is not used by agents; the browser uses gRPC-Web streaming) [04_P1 §3].
- Enroll, AdvertisePrefixes, ReportStatus are **unary**.
- Compression: the server enables gzip for unary responses; WatchNetworkMap frames are small (deltas), so per-frame compression is **off by default** to keep p99 latency low (compression CPU on tiny frames is net-negative).
UNVERIFIED: the exact frame-size threshold below which compression hurts is to be measured during the §5 soak; until measured, off-by-default stands.
- Idle-timeout: the server sends a KeepAlive every **20 s** ([research-go_cp §6.3] ticker). The agent treats **2 missed keepalives (~45 s wall, allowing jitter)** as a dead stream and reconnects with its last known_version (§3.3).
UNVERIFIED: 45 s is a design default, not yet load-tuned.

3. WatchNetworkMap semantics — the state machine

WatchNetworkMap is the contract's spine. It delivers desired state as **one snapshot then a delta stream**, with cheap reconnect via `known_version`. Peers in every frame are **already policy-filtered** (C4): the wire never carries a peer the receiving node may not reach.

3.1 Server-side stream lifecycle (Go signature + sketch)

```
// internal/coordinator/watch.go
//
// WatchNetworkMap is the Connect server handler. It authenticates
// the device from the
// mTLS cert (§4.4), registers an in-memory subscription, sends
// the initial snapshot OR
// catch-up deltas (§3.3), then forwards minimal deltas +
// keepalives until disconnect.
func (c *Coordinator) WatchNetworkMap(
    ctx context.Context,
    req *connect.Request[coordinatorv1.WatchRequest],
    stream *connect.ServerStream[coordinatorv1.MapUpdate],
) error {
    dev, err := authDevice(ctx)           // mTLS cert ->
    device row (§4.4)
    if err != nil {
        return err                       // already a
        *connect.Error (Unauthenticated)
    }
    if req.Msg.DeviceId != dev.ID.String() { // body must
        agree with cert
        return connect.NewError(connect.CodePermissionDenied,
            errDeviceMismatch)
    }
    if dev.RevokedAt.Valid {                // belt-and-
        suspenders: a revoked device never streams
        return connect.NewError(connect.CodePermissionDenied,
            errRevoked)
    }

    sub := c.subscribe(dev.TenantID, dev.ID) // registers
    stream; flips presence ONLINE; emits device.online
    defer sub.Close()                       // flips
    presence OFFLINE; emits device.offline

    cur := c.version(dev.TenantID)          // current
    tenant map version
    switch {
    case req.Msg.KnownVersion == 0 || req.Msg.KnownVersion > cur
        || !c.canResume(dev, req.Msg.KnownVersion):
        // full snapshot path (§3.3 case A)
        if err := stream.Send(snapshotUpdate(cur,
            c.buildMap(dev))); err != nil {
            return err
        }
    }
```

```

default:
    // resume path: replay compacted deltas (cur >=
    // known_version) (§3.3 case B)
    for _, d := range c.deltasSince(dev,
    req.Msg.KnownVersion) {
        if err := stream.Send(deltaUpdate(d)); err != nil {
            return err
        }
    }
}

ka := time.NewTicker(20 * time.Second)
defer ka.Stop()
for {
    select {
    case <-ctx.Done():
        return
        ctx.Err() // client gone / server
        shutdown
    case d, ok := <-sub.C:
        if !ok
        { // back-pressure drop
            (§3.5): force reconnect-with-snapshot
            return
            connect.NewError(connect.CodeResourceExhausted,
            errSlowConsumer)
        }
        if err := stream.Send(deltaUpdate(d)); err != nil {
            return err
        }
    case <-ka.C:
        if err :=
        stream.Send(keepAliveUpdate(c.version(dev.TenantID)));
        err != nil {
            return err
        }
    }
}
}

```

Helper constructors keep the oneof invariant (exactly one body set):

```

func snapshotUpdate(v uint64, m *coordinatorv1.NetworkMap)
    *coordinatorv1.MapUpdate {
    return &coordinatorv1.MapUpdate{Version: v,
    Body: &coordinatorv1.MapUpdate_Snapshot{Snapshot: m}}
}

func deltaUpdate(d Delta) *coordinatorv1.MapUpdate {
    return &coordinatorv1.MapUpdate{Version: d.Version,
    Body: &coordinatorv1.MapUpdate_Delta{Delta: d.Proto()}}
}

func keepAliveUpdate(v uint64) *coordinatorv1.MapUpdate {
    return &coordinatorv1.MapUpdate{Version: v,

```

```

        Body: &coordinatorv1.MapUpdate_Keepalive{Keepalive:
              &coordinatorv1.KeepAlive{}}}}
    }

```

3.2 Stream state machine (agent's view)

```

stateDiagram-v2
    [*] --> Connecting: open WatchNetworkMap(device_id,
known_version)
    Connecting --> Authenticating: TCP+TLS+mTLS established
    Authenticating --> Snapshotting: cert valid, device not
revoked
    Authenticating --> [*]: Unauthenticated / PermissionDenied
($4)
    Snapshotting --> Live: received MapUpdate{snapshot} OR catch-
up deltas
    Live --> Live: MapUpdate{delta} (apply: upsert/remove peers,
maybe transport/dns)
    Live --> Live: MapUpdate{keepalive} (reset liveness timer)
    Live --> Reconnect: 2 missed keepalives OR transport error OR
ResourceExhausted
    Reconnect --> Connecting: reopen with last applied version as
known_version
    Live --> [*]: PermissionDenied (revoked mid-stream, §4.7)

```

Agent reconciliation rules (normative). On each frame the agent:

1. snapshot → discard all prior peer state, install NetworkMap wholesale, set local appliedVersion = MapUpdate.version.
2. delta → for each upsert_peers[i] add-or-replace the peer keyed by device_id; for each remove_peer_ids[j] delete that peer; if transport is set, replace transport policy; if dns is set, replace DNS config; set appliedVersion = MapUpdate.version. An unset transport/dns means **unchanged** (do not clear them).
3. keepalive → reset the liveness timer; do not change appliedVersion (the version field on a keepalive equals the current tenant version and is informational).

3.3 known_version cheap-reconnect protocol

known_version lets an agent that briefly disconnected resume **without a full resync** [04_P1 §3, research-go_cp §4].

```

sequenceDiagram
    autonumber
    participant Ag as Agent
    participant Co as Coordinator
    Note over Ag: first ever connect
    Ag->>Co: WatchNetworkMap(device_id, known_version=0)
    Co-->>Ag: MapUpdate{version=42, snapshot=...}
    Note over Ag: appliedVersion=42, stream drops at 47
    Ag->>Co: WatchNetworkMap(device_id, known_version=42)
    alt deltas 43..47 retained (case B: resume)

```

```

    Co-->>Ag: MapUpdate{version=43, delta}
    Co-->>Ag: MapUpdate{version=47, delta}
    else delta history evicted / known_version > current / device's
visibility changed (case A: resync)
    Co-->>Ag: MapUpdate{version=47, snapshot=...}
end

```

Server decision (the switch in §3.1).

- **Case A — full snapshot** when ANY of: `known_version == 0`;
`known_version > cur` (client claims a future version — clock/replay anomaly,
resync defensively); `!canResume` (the compacted delta history for this tenant no
longer reaches back to `known_version`, OR the device's *visibility set* changed
in a way a delta cannot express safely — e.g. the device was moved between
groups so peers it must now FORGET overlap with peers it must now LEARN;
resync is the correct conservative action per C4 need-to-know).
- **Case B — catch-up deltas** otherwise: replay every retained delta with version
in `(known_version, cur]` in ascending version order, then enter the live loop.

Delta retention. The coordinator keeps a bounded **per-tenant ring buffer** of the
last `K` compacted deltas (default `K = 256` versions; a design default, not yet tuned —
UNVERIFIED the optimal `K`). If `known_version < cur - K`, the history cannot
serve the resume → Case A. The ring is in-memory only (R2: coordinator owns no
durable tables); after a coordinator restart the ring is empty, so the first reconnect
from every agent is a Case-A snapshot — acceptable because the graph itself is
rehydrated from Postgres + events on boot [research-go_cp §6.1].

3.4 Policy-prefiltering guarantee (C4 on the wire)

Every Peer in a `NetworkMap.peers` or `MapDelta.upsert_peers` is already the
output of the policy compiler's `VisibleTo/AllowedIPs` for **this** device [research-
go_cp §6.2/§7.2]:

- A device NEVER receives a Peer it has no compiled rule to reach. "Need-to-know" is enforced at map-build time, not by the client filtering. This is a privacy invariant (C4) and a security invariant simultaneously: a compromised client learns only its authorized peers.
- `Peer.allowed_ips` is the **coarse, CIDR-only** WG AllowedIPs (WireGuard cannot express ports). The **fine, port-level** verdicts live in the edge's nftables/eBPF verdict map (Volume 1), not on this wire — the proto carries exactly what the data plane's WG layer needs, no more.
- When a policy change REMOVES a device's visibility of a peer, the coordinator emits a `MapDelta.remove_peer_ids` entry (not a snapshot, unless §3.3 Case A triggers) so the client drops the WG peer immediately.

3.5 Back-pressure (slow-consumer safety)

Each subscription has a **bounded send queue** (`sub.C`, default capacity **64** deltas). If the queue fills (a wedged/slow agent), the coordinator closes `sub.C`; the handler returns `ResourceExhausted` (§4.2); the agent reconnects with its `known_version` and gets a fresh snapshot or catch-up. This bounds coordinator memory under a slow consumer — it never grows the queue unboundedly (the §5 24 h soak asserts zero memory growth) [research-go_cp §6.4].

4. Authz, error taxonomy, and edge cases

4.1 Enroll — the only unauthenticated RPC

Enroll carries no mTLS cert (the device does not yet have one). It authenticates by a **single-use, short-lived enroll token** minted by Console/identity and stored **hashed** [research-go_cp §9.1].

Server algorithm (atomic):

```
// internal/identity/enroll.go (sketch – full impl in research-
// go_cp §9.2)
func (s *Identity) Enroll(ctx context.Context,
    req *connect.Request[coordinatorv1.EnrollRequest],
) (*connect.Response[coordinatorv1.EnrollResponse], error) {
    m := req.Msg
    if err := validateEnroll(m); err != nil {    // §4.6 input
        validation
        return nil, err
    }
    // verify+consume token in ONE tx; a replay of a consumed
    // token fails here.
    tok, err := s.consumeToken(ctx, hash(m.EnrollToken))
    if err != nil {
        return nil, connect.NewError(connect.CodeUnauthenticated,
            errBadOrUsedToken)
    }
    return s.store.WithTenant(ctx, tok.TenantID, func(q
        *db.Queries) (...) {
        ip, _ := s.ipam.AllocOverlayIP(ctx, q,
            tok.TenantID)    // D4
        cert, _ := s.pki.IssueDeviceCert(ctx, deviceID,
            24*time.Hour) // mTLS
        // INSERT device row; emit device.enrolled on the bus (R3)
    })
}
```

- Token is **single-use**: consumeToken deletes/marks it inside the same tx, so two concurrent enrolls with the same token: one wins, the other gets Unauthenticated.
- Token TTL is short (**UNVERIFIED**: exact TTL — design default 15 min; operator-configurable).
- wg_pubkey MUST be exactly 32 bytes (Curve25519). A wrong length → InvalidArgument (§4.6).

4.2 Connect error-code taxonomy (normative)

Every failure maps to a Connect/gRPC code so all three generated stubs surface it identically. The table is the contract: a server returning a different code for the same condition is a defect.

Condition	Connect code	Sentinel	Retriable by agent?
Enroll token bad / already used / expired	Unauthenticated	errBadOrUsedToken	No (mint a new token)
Missing/invalid mTLS cert on an authed RPC	Unauthenticated	errNoCert	No (re-enroll)
wg_pubkey not 32 bytes / empty token / DEVICE_KIND_UNSPECIFIED	InvalidArgument	errBadField	No (fix input)
WatchRequest.device_id $\neq$ cert's device	PermissionDenied	errDeviceMismatch	No
Device revoked (at open OR mid-stream)	PermissionDenied	errRevoked	No (re-enroll)
AdvertisePrefixes from a non-connector device	PermissionDenied	errNotConnector	No
Advertised CIDR malformed / overlaps reserved	InvalidArgument	errBadCIDR	No (fix CIDRs)
Slow consumer, send queue full	ResourceExhausted	errSlowConsumer	Yes (reconnect w/ known_version)
Coordinator graph not yet hydrated (boot)	Unavailable	errNotReady	Yes (backoff + retry)
Postgres/Redis transient failure	Unavailable	errBackendDown	Yes (backoff)
Per-device stream cap exceeded	ResourceExhausted	errTooManyStreams	Yes (backoff)
Context cancelled / deadline	Canceled / DeadlineExceeded	—	Yes

Retry discipline. Agents retry only Unavailable/ResourceExhausted/Canceled/DeadlineExceeded with exponential backoff + jitter (base 500 ms, cap 30 s — **UNVERIFIED:** exact schedule, design default). They never retry Unauthenticated/PermissionDenied/InvalidArgument (those need human/operator action — re-enroll or fix input).

4.3 Authz matrix per RPC

RPC	AuthN	AuthZ predicate
Enroll	hashed enroll token	token valid + unused + unexpired; tenant resolved from token
WatchNetworkMap	device mTLS cert	cert→device row; not revoked; device_id matches cert; tenant = cert's tenant
AdvertisePrefixes	device mTLS cert	as above AND device.kind == CONNECTOR
ReportStatus		

RPC	AuthN	AuthZ predicate
	device	cert→device row; not revoked;
	mTLS cert	device_id matches cert

authDevice(ctx) resolves the **cert serial** → **device_certs** row → **devices** row, rejecting on revoked/expired [research-go_cp §8.2]:

```
// internal/api/authdevice.go
func authDevice(ctx context.Context) (registry.Device, error) {
    tlsState, ok := connectTLSFrom(ctx)           // peer cert
    from the Connect/HTTP2 conn
    if !ok || len(tlsState.PeerCertificates) == 0 {
        return registry.Device{},
            connect.NewError(connect.CodeUnauthenticated, errNoCert)
    }
    serial := tlsState.PeerCertificates[0].SerialNumber.String()
    dev, err := lookupBySerial(ctx, serial)        //
    device_certs JOIN devices, WHERE NOT revoked
    if err != nil {
        return registry.Device{},
            connect.NewError(connect.CodeUnauthenticated, errNoCert)
    }
    return dev, nil
}
```

4.4–4.7 Edge cases (each is a test point, §7)

- **§4.4 cert/body mismatch.** A device presenting cert for D1 but sending WatchRequest{device_id: D2} → PermissionDenied (errDeviceMismatch). Prevents one valid device impersonating another.
- **§4.5 AdvertisePrefixes is declarative, not a diff.** cidrs is the **complete** set the connector serves; the server diffs against stored prefixes and emits connector.prefixes.changed only if the set actually changed (idempotent re-send is a no-op, accepted=true, conflicts=[]). Overlap with another connector's CIDR does NOT reject (returns accepted=true + a conflicts[] advisory) — the overlap is resolved by 4via6 site disambiguation (D4), surfaced in Console via a route.conflict.detected event [research-go_cp §7.3]. A **malformed** CIDR DOES reject (InvalidArgument, errBadCIDR).
- **§4.6 input validation (fail-closed).** Enroll: non-empty token; wg_pubkey length == 32; kind ∈ {CLIENT, CONNECTOR} (UNSPECIFIED rejected); os must be one of the seven known strings ([research-go_cp DDL devices.os comment]) — an unknown os is rejected InvalidArgument rather than stored, so the DB never holds an unrecognized platform tag.
- **§4.7 revoke mid-stream.** When device.revoked fires for a device with an open WatchNetworkMap, the coordinator's event handler closes that subscription; the handler returns PermissionDenied (errRevoked); the edge drops the WG peer. Target: stream force-closed within the convergence SLO (< 1 s, §5) [research-go_cp §9.3].

4.8 ReportStatus no-logging guarantee (C3 on the wire)

StatusReport is defined to carry **only** device_id, transport, rtt_ms. The proto has no field for bytes/flows/destinations, so the contract makes traffic-logging **structurally impossible** at the wire layer — mirroring the Postgres no-logging guarantee [research-go_cp §2.4]. A meta-test (§7) asserts the StatusReport descriptor has exactly these three fields; adding a bytes_total field **MUST** make that test FAIL (paired §1.1 mutation). This is the wire-level runtime signature (§11.4.108) that proves C3, not merely promises it.

5. Convergence SLO (the < 1 s promise) tied to this contract

The contract exists to make desired-state propagation **fast and measurable**. The binding SLO for the streaming path [research-go_cp §10.2, 04_P1 §10]:

Metric	Target	Measured at the contract boundary
event → stream.Send(MapDelta) on wire	p99 < 1 s	coordinator histogram helix_reconcile_seconds, from event-receive to the stream.Send return for the affected WatchNetworkMap
device revoke → stream force-closed + edge WG peer removed	< 1 s	revoke timer; the errRevoked return + edge confirmation
enrollment round-trip (Enroll unary)	< 500 ms	Enroll server histogram
coordinator RSS @ 10k open streams	bounded, ~0 slope over 24 h	process_resident_memory_bytes

sequenceDiagram

autonumber

participant Adm as Console

participant Pol as policy

participant Bus as Redis Streams

participant Co as coordinator

participant Ag as Agent (open WatchNetworkMap)

Adm->>Pol: activate policy v

Pol->>Bus: XADD events:policy {policy.compiled, v}

Bus-->>Co: XReadGroup delivers (start helix_reconcile_seconds timer)

Co-->>Co: recompute minimal affected set (§3.4)

Co->>Ag: stream.Send(MapUpdate{version=v, delta}) (stop timer)

Note over Adm,Ag: p99 < 1s, event->delta-on-wire, zero restarts

The < 1 s number is an **anti-bluff acceptance number** (§11.4.5/§11.4.69): it is asserted with captured evidence from `helix_reconcile_seconds`, not assumed. A green functional test that does not also assert the p99 histogram is incomplete for this contract.

6. buf codegen pipeline (Go / Dart / Rust)

One `.proto`, three generated stubs, no hand-written clients — so the Go server, Dart client core (via `flutter_rust_bridge` boundary), and Rust edge/connector cannot drift [04_ARCH §4.2, research-go_cp §4]. The proto lives in `helix-proto` (`proto/helix/coordinator/v1/coordinator.proto`) [SYNTHESIS §6].

6.1 buf.yaml (lint + breaking config)

```
# helix-proto/buf.yaml
version: v2
modules:
  - path: proto
lint:
  use:
    - STANDARD
  except:
    - PACKAGE_VERSION_SUFFIX # we use helix.coordinator.v1
                             (suffix present) – kept; documented here
breaking:
  use:
    - FILE # field-number/type changes within
           v1 are breaking -> CI gate (§6.3)
```

6.2 buf.gen.yaml (three language outputs)

```
# helix-proto/buf.gen.yaml
version: v2
plugins:
  # ---- Go (server) ----
  - remote: buf.build/protocolbuffers/go
    out: gen
    opt: paths=source_relative
  - remote: buf.build/connectrpc/go # Connect Go handlers/
    clients
    out: gen
    opt: paths=source_relative
  # ---- Dart (client core boundary) ----
  - remote: buf.build/protocolbuffers/dart
    out: gen/dart
  # ---- Rust (edge / connector) ----
  - remote: buf.build/community/neoeinstein-prost # prost
    messages
    out: gen/rust/src
```

```

- remote: buf.build/community/neoeinstein-tonic    # tonic
  service stubs
  out: gen/rust/src

```

UNVERIFIED: the exact remote plugin coordinates/versions above are the conventional buf plugin names; the pinned versions are chosen at implementation time and recorded in buf.gen.yaml then — they are not asserted here as fixed.

6.3 Generation + breaking-change gate (runs in the local pre-build, §11.4.156 NO CI)

```

buf lint                                # STANDARD rules
buf breaking --against '.git#branch=main' # field renumber/type-
  change -> FAIL (protects v1 wire)
buf generate                            # emits gen/ (Go), gen/dart,
  gen/rust

```

The buf breaking gate is the mechanical enforcement that the §1.1 field-number registry stays frozen: any commit renumbering a v1 field, changing a field type, or reusing a removed number fails the gate. Paired §1.1 mutation: a test branch that renames Peer.allowed_ips field number 3 → 7 MUST make buf breaking FAIL; reverting MUST make it pass.

7. Test points (tied to constitution §11.4.169 comprehensive test-type coverage)

§11.4.169 mandates comprehensive coverage across every applicable test type for this contract. Each row below is an implementation-ready test point; every PASS cites captured evidence (§11.4.5/§11.4.69/§11.4.107) — a green test that does not exercise the wire is a §11.4 PASS-bluff.

#	Test type (§11.4.169)	Target	Concrete assertion + evidence
T1	unit	oneof body constructors	each of snapshotUpdate/deltaUpdate/keepAliveUpdate sets exactly one body variant; round-trip proto.Marshal/Unmarshal is identity
T2	unit	C3 wire guarantee (§4.8)	reflection over StatusReport descriptor: field set == {device_id, transport, rtg_ms}; mutation add bytes_total → test FAILs (§1.1)
T3	unit	field-number registry (§1.1)	golden descriptor snapshot; any number change diffs → FAIL
T4	unit	input validation (§4.6)	table-driven: 31-byte / 33-byte wg_pubkey, empty token, DEVICE_KIND_UNSPECIFIED, unknown os → InvalidArgument
T5	integration	Enroll happy path	spin Postgres+Redis via vasic-digital/containers (§11.4.76, not ad-hoc docker); Enroll → row inserted, device.enrolled

#	Test type (§11.4.169)	Target	Concrete assertion + evidence
			on bus, response has 32-byte-derived cert + ULA overlay_ip
T6	integration	known_version resume (§3.3)	connect w/ known_version=0 (snapshot), drop, reconnect w/ last version: assert delta delivered (Case B); evict ring → assert snapshot (Case A)
T7	integration	policy-prefiltering (§3.4)	two devices, ACL grants only one a connector; assert the ungranted device's stream NEVER carries that Peer (need-to-know, C4)
T8	integration	revoke mid-stream (§4.7)	open WatchNetworkMap, revoke device, assert stream returns PermissionDenied(errRevoked) within < 1 s; captured timer
T9	e2e / full-automation	enroll→advertise→policy→delta	drive Enroll(connector)+Enroll(client) → AdvertisePrefixes → activate policy → assert MapDelta content on the client stream + helix_reconcile_seconds p99 < 1 s (captured histogram)
			non-connector calling AdvertisePrefixes →
T10	security	authz matrix (§4.3)	PermissionDenied(errNotConnector); missing cert → Unauthenticated; cert/body mismatch → PermissionDenied(errDeviceMismatch)
T11	security	enroll-token replay (§4.1)	two concurrent Enroll with same token: exactly one succeeds, other Unauthenticated; consumed token re-use → Unauthenticated
T12	fuzz	wire robustness	fuzz Unmarshal of WatchRequest/EnrollRequest/AdvertiseRequest with random/truncated bytes; assert no panic, all malformed inputs → InvalidArgument, never a crash (§11.4.1 FAIL-bluff guard)
T13	stress	fan-out scale	N=10k simulated agents holding streams; flap a tenant policy; assert all affected streams receive the delta + p99 < 1 s; non-affected streams receive nothing
T14	chaos	slow consumer (§3.5)	wedge one agent's reader; assert its sub.C fills, it gets ResourceExhausted, reconnects with snapshot; coordinator RSS does NOT grow (other streams unaffected — single-resource isolation)
T15	chaos	backend flap	kill Redis mid-stream; assert open streams keep their last state (fail-static, C1), coordinator returns Unavailable on new connects, recovers on Redis return; reclaimed PEL entries not lost (§11.4.147)

#	Test type (§11.4.169)	Target	Concrete assertion + evidence
T16	performance / soak	24 h memory	10k streams, periodic policy flaps for 24 h; assert <code>process_resident_memory_bytes</code> slope ≈ 0 (bounded send queues, ring eviction)
T17	contract / compat	buf breaking (§6.3)	renumber <code>Peer.allowed_ips</code> → buf breaking FAILs; revert → passes (paired §1.1 mutation)
T18	contract / codegen	cross-lang stubs	buf generate emits Go+Dart+Rust; a Rust tonic client and a Go server complete one <code>WatchNetworkMap</code> round-trip (prevents stub drift)
T19	challenge (HelixQA)	end-user reachability	a challenges entry drives <code>enroll</code> → <code>policy</code> → <code>WatchNetworkMap</code> → <code>edge reconcile</code> and asserts the authorized LAN host is reachable AND an unauthorized one is denied, with captured evidence (the MVP DoD slice)

Integration/e2e infra is booted **on demand via vasic-digital/containers** (§11.4.76) — never ad-hoc `docker run`; QA/anti-bluff via `helix_qa` + challenges (§11.4.27/§11.4.5/.69/.107) [SYNTHESIS §8].

8. Forward seams (this contract extends without a v1 break) [04_P1 §12]

Phase-2 additions are **field additions** (new field numbers, backward-compatible) or new RPCs, never v1 rennumbers — so buf breaking stays green:

- `TransportPolicy.order` gains `shadowsocks/udp-over-tcp/daita` ladder entries (string values, no schema change).
- `Peer.endpoint` is joined by a future repeated `PeerCandidate` `direct_candidates = 7`; for direct P2P NAT-traversal so traffic stops always relaying through the gateway (new field number 7).
- `GatewayInfo` gains a future multi-hop path field for nested-WG multi-hop.
- `EnrollResponse/Peer` gain a future PQ pre-shared-key field for the post-quantum handshake.

Each is additive because v1 drew the field-number registry (§1.1) with room and froze it under the buf breaking gate (§6.3). The byte-level rules that make these additions safe across live agents are the **WatchNetworkMap contract** document (this Volume) — referenced, not duplicated here.

9. Sources

- **[research-go_cp]** docs/research/mvp/final/02-control-plane.md — §4 (proto), §6 (coordinator stream loop), §7 (policy compiler outputs the proto carries), §8 (Connect binding + authz), §9 (enrollment/PKI), §10 (SLOs + test strategy), §0.1 (C-rules).
- **[04_P1]** docs/research/mvp/04_VPN_CLD/HelixVPN-Phase1-MVP.md — §3 agent contract (proto origin, known_version, snapshot-then-deltas, policy-prefiltering), §4 coordinator, §5 events, §6 enrollment, §10 SLO, §11 MVP DoD.
- **[04_ARCH §N]** docs/research/mvp/04_VPN_CLD/HelixVPN-Architecture-Refined.md — §3.4 ULA /48 + 4via6 (D4), §4.2 schema-generated clients (no hand-written stubs), §7 no-logging/default-deny.
- **[SYNTHESIS §N]** v09-research/_SYNTHESIS.md — §2 stack floor, §3 decisions D1–D7, §6 repo layout (helix-proto), §7 security invariants, §8 ecosystem submodule wiring, §9 constitution bindings.
- Constitution anchors cited: §11.4.5/§11.4.69/§11.4.107 (captured-evidence/anti-bluff), §11.4.6 (no-guessing — UNVERIFIED marks), §11.4.27 (test-type coverage), §11.4.35 (visible reconciliation), §11.4.76 (containers submodule), §11.4.108 (runtime-signature), §11.4.147 (no-work-loss), §11.4.156 (no active CI), §11.4.169 (comprehensive test-type coverage), §1.1 (paired meta-test mutations).